



Juergen Gall

Structure from motion MA-INF 2201 / MA-MOROB-M04 WS25/26

Exam

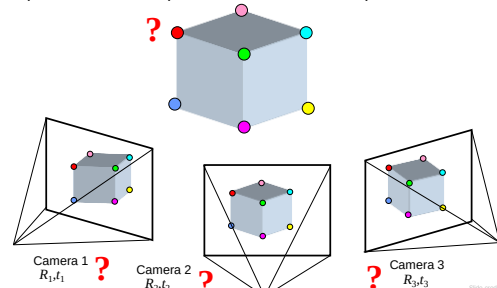
- Recap lecture: 6.2.
- 1st Exam: 24.02.2026: 12:30-14:30
Lecture hall 2, Friedrich-Hirzebruch-Allee 5, Hörsaalzentrum
- 2nd Exam: 20.03.2026: 9:30-11:30
Lecture hall 2, Friedrich-Hirzebruch-Allee 5, Hörsaalzentrum

Seminar

- MA-INF 2206 - Seminar Vision
- Meetings:
 - Today, 3.2., 11:45
HS XVI, Nußallee 17
 - Monday, 9.2., 11:45
2.025, Institute of Computer Science, Friedrich-Hirzebruch-Allee 8

Structure from motion

Given a set of corresponding points in two or more images, compute the camera parameters and the 3D point coordinates

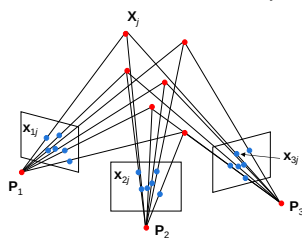


Structure from motion

Given: m images of n fixed 3D points

$$\mathbf{x}_{ij} = \mathbf{P}_i \mathbf{X}_j, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

Problem: estimate m projection matrices $\mathbf{P}_i$ and n 3D points $\mathbf{X}_j$ from the mn correspondences $\mathbf{x}_{ij}$



Structure from motion ambiguity

- If we scale the entire scene by some factor k and, at the same time, scale the camera matrices by the factor of $1/k$, the projections of the scene points in the image remain exactly the same:

$$\mathbf{x} = \mathbf{P}\mathbf{X} = \left(\frac{1}{k} \mathbf{P} \right) (k \mathbf{X})$$

It is impossible to recover the absolute scale of the scene!

Structure from motion ambiguity



If we scale the entire scene by some factor k and, at the same time, scale the camera matrices by the factor of $1/k$, the projections of the scene points in the image remain exactly the same

More generally: if we transform the scene using a transformation Q and apply the inverse transformation to the camera matrices, then the images do not change

$$\mathbf{x} = \mathbf{P}\mathbf{X} = (\mathbf{P}\mathbf{Q}^{-1})(\mathbf{Q}\mathbf{X})$$

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Types of ambiguity



Projective
15dof

$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^T & v \end{bmatrix}$$



Preserves intersection and tangency

Affine
12dof

$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$



Preserves parallelism, volume ratios

Similarity
7dof

$$\begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$



Preserves angles, ratios of length

Euclidean
6dof

$$\begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$



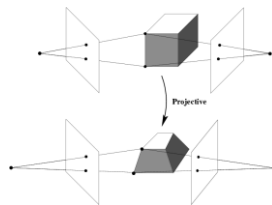
Preserves angles, lengths

- With no constraints on the camera calibration matrix or on the scene, we get a *projective* reconstruction
- Need additional information to *upgrade* the reconstruction to affine, similarity, or Euclidean

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Projective ambiguity



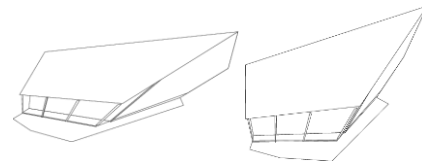
$$\mathbf{Q}_P = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^T & v \end{bmatrix}$$

$$\mathbf{x} = \mathbf{P}\mathbf{X} = (\mathbf{P}\mathbf{Q}_P^{-1})(\mathbf{Q}_P\mathbf{X})$$

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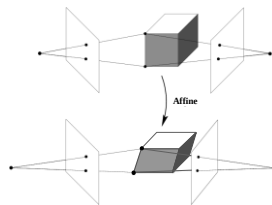
Projective ambiguity



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Affine ambiguity



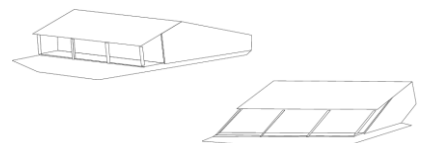
$$\mathbf{Q}_A = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

$$\mathbf{x} = \mathbf{P}\mathbf{X} = (\mathbf{P}\mathbf{Q}_A^{-1})(\mathbf{Q}_A\mathbf{X})$$

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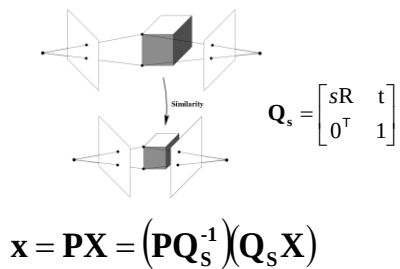
Affine ambiguity



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Similarity ambiguity

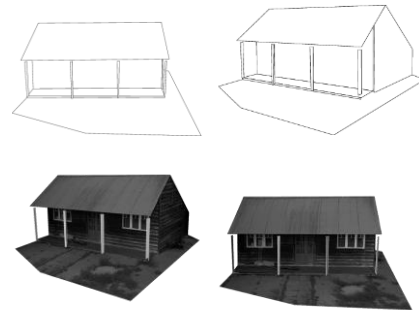


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Similarity ambiguity



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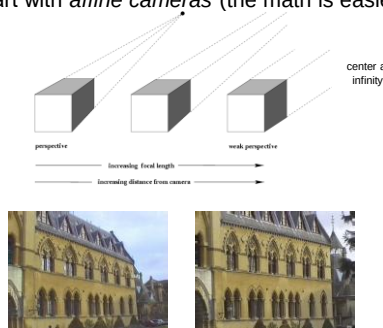
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Structure from motion



- Let's start with *affine cameras* (the math is easier)



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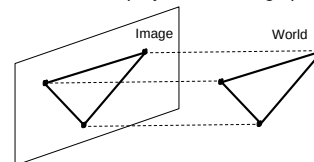
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Recall: Orthographic Projection



Special case of perspective projection

- Distance from center of projection to image plane is infinite



- Projection matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \Rightarrow (x, y)$$

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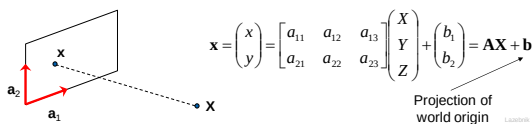
Affine cameras



- A general affine camera combines the effects of an affine transformation of the 3D space, orthographic projection, and an affine transformation of the image:

$$\mathbf{P} = [\mathbf{3} \times \mathbf{3} \text{ affine}] \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} [\mathbf{4} \times \mathbf{4} \text{ affine}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0} & 1 \end{bmatrix}$$

- Affine projection is a linear mapping + translation in inhomogeneous coordinates



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Affine structure from motion



- Given: m images of n fixed 3D points:

$$\mathbf{x}_{ij} = \mathbf{A}_i \mathbf{X}_j + \mathbf{b}_i, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$
- Problem: use the mn correspondences $\mathbf{x}_{ij}$ to estimate m projection matrices $\mathbf{A}_i$ and translation vectors $\mathbf{b}_i$, and n points $\mathbf{X}_j$
- The reconstruction is defined up to an arbitrary affine transformation $\mathbf{Q}$ (12 degrees of freedom):

$$\begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0} & 1 \end{bmatrix} \rightarrow \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{Q}^{-1}, \quad \begin{pmatrix} \mathbf{x} \\ 1 \end{pmatrix} \rightarrow \mathbf{Q} \begin{pmatrix} \mathbf{x} \\ 1 \end{pmatrix}$$

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Recall: Types of ambiguity

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Affine structure from motion

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- Given: m images of n fixed 3D points:
 $\mathbf{x}_{ij} = \mathbf{A}_i \mathbf{X}_j + \mathbf{b}_i, \quad i = 1, \dots, m, j = 1, \dots, n$
- Problem: use the mn correspondences $\mathbf{x}_{ij}$ to estimate m projection matrices $\mathbf{A}_i$ and translation vectors $\mathbf{b}_i$, and n points $\mathbf{X}_j$
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$$\begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0} & 1 \end{bmatrix} \rightarrow \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{Q}^{-1}, \quad \begin{pmatrix} \mathbf{x} \\ 1 \end{pmatrix} \rightarrow \mathbf{Q} \begin{pmatrix} \mathbf{x} \\ 1 \end{pmatrix}$$

- We have $2mn$ knowns and $8m + 3n$ unknowns (minus 12 dof for affine ambiguity)
- Thus, we must have $2mn \geq 8m + 3n - 12$
- For two views, we need four point correspondences

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Affine structure from motion

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- Centering: subtract the centroid of the image points

$$\hat{\mathbf{x}}_{ij} = \mathbf{x}_{ij} - \frac{1}{n} \sum_{k=1}^n \mathbf{x}_{ik} = \mathbf{A}_i \mathbf{X}_j + \mathbf{b}_i - \frac{1}{n} \sum_{k=1}^n (\mathbf{A}_i \mathbf{X}_k + \mathbf{b}_i)$$

$$= \mathbf{A}_i \left(\mathbf{X}_j - \frac{1}{n} \sum_{k=1}^n \mathbf{X}_k \right) = \mathbf{A}_i \hat{\mathbf{X}}_j$$

- For simplicity, assume that the origin of the world coordinate system is at the centroid of the 3D points
- After centering, each normalized point $\mathbf{x}_{ij}$ is related to the 3D point $\mathbf{X}_j$ by

$$\hat{\mathbf{x}}_{ij} = \mathbf{A}_i \mathbf{X}_j$$

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Affine structure from motion

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- Let's create a $2m \times n$ data (measurement) matrix:

$$\mathbf{D} = \begin{bmatrix} \hat{\mathbf{x}}_{11} & \hat{\mathbf{x}}_{12} & \cdots & \hat{\mathbf{x}}_{1n} \\ \hat{\mathbf{x}}_{21} & \hat{\mathbf{x}}_{22} & \cdots & \hat{\mathbf{x}}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\mathbf{x}}_{m1} & \hat{\mathbf{x}}_{m2} & \cdots & \hat{\mathbf{x}}_{mn} \end{bmatrix}$$

points (n)

cameras ($2m$)

C. Tomasi and T. Kanade. Shape and motion from image streams under orthography: A factorization method. *IJCV*, 9(2):137-154, November 1992.

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cameras ($2m \times 3$)

points ($3 \times n$)

The measurement matrix $\mathbf{D} = \mathbf{MS}$ must have rank 3!

C. Tomasi and T. Kanade. Shape and motion from image streams under orthography: A factorization method. *IJCV*, 9(2):137-154, November 1992.

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Factorizing the measurement matrix

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$$\begin{matrix} \text{Measurements} & = & \text{Motion} & \times & \text{Shape} \end{matrix}$$

$\mathbf{D} = \mathbf{MS}$

Measurements: $2m \times n$
Motion: 3×3
Shape: $3 \times n$

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Factorizing the measurement matrix

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- Singular value decomposition of D :

$$\begin{array}{c}
 \begin{array}{c} n \\ 2m \end{array} D = \begin{array}{c} n \\ U \end{array} \times \begin{array}{c} n \\ W \end{array} \times \begin{array}{c} n \\ V^T \end{array} \\
 \\
 \begin{array}{c} 2m \\ D \end{array} = \begin{array}{c} 3 \\ U_3 \end{array} \times \begin{array}{c} 3 \\ W_3 \end{array} \times \begin{array}{c} 3 \\ V_3^T \end{array}
 \end{array}$$

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Factorizing the measurement matrix

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- Singular value decomposition of D :

$$\begin{array}{c}
 \begin{array}{c} n \\ 2m \end{array} D = \begin{array}{c} n \\ U \end{array} \times \begin{array}{c} n \\ W \end{array} \times \begin{array}{c} n \\ V^T \end{array} \\
 \\
 \begin{array}{c} 2m \\ D \end{array} = \begin{array}{c} 3 \\ U_3 \end{array} \times \begin{array}{c} 3 \\ W_3 \end{array} \times \begin{array}{c} 3 \\ V_3^T \end{array}
 \end{array}$$

To reduce to rank 3, we just need to set all the singular values to 0 except for the first 3

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Factorizing the measurement matrix

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- Obtaining a factorization from SVD:

$$\begin{array}{c}
 \begin{array}{c} 2m \\ D \end{array} = \begin{array}{c} 3 \\ U_3 \end{array} \times \begin{array}{c} 3 \\ W_3 \end{array} \times \begin{array}{c} n \\ V_3^T \end{array}
 \end{array}$$

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Factorizing the measurement matrix

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- Obtaining a factorization from SVD:

$$\begin{array}{c}
 \begin{array}{c} 2m \\ D \end{array} = \begin{array}{c} 3 \\ U_3 \end{array} \times \begin{array}{c} 3 \\ W_3 \end{array} \times \begin{array}{c} n \\ V_3^T \end{array}
 \end{array}$$

Possible decomposition:
 $M = U_3 W_3^{1/2} \quad S = W_3^{1/2} V_3^T$

$$\begin{array}{c} D \end{array} = \begin{array}{c} M \end{array} \times \begin{array}{c} S \end{array}$$

This decomposition minimizes $\|D - MS\|^2$

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Affine ambiguity

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$$\begin{array}{c} D \end{array} = \begin{array}{c} M \end{array} \times \begin{array}{c} S \end{array}$$

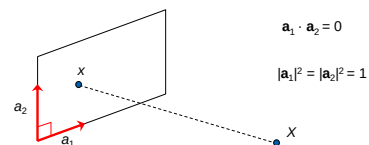
- The decomposition is not unique. We get the same D by using any 3×3 matrix C and applying the transformations $M \rightarrow MC$, $S \rightarrow C^{-1}S$
- That is because we have only an affine transformation and we have not enforced any Euclidean constraints (like forcing the image axes to be perpendicular, for example)

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Eliminating the affine ambiguity

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- Orthographic: image axes are perpendicular and of unit length



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Solve for orthographic constraints



- Three equations for each image i

$$\begin{aligned} \mathbf{m}_{i1}^T \mathbf{C} \mathbf{C}^T \mathbf{m}_{i1} &= 1 \\ \mathbf{m}_{i2}^T \mathbf{C} \mathbf{C}^T \mathbf{m}_{i2} &= 1 \quad \text{where} \quad \mathbf{M}_i = \begin{bmatrix} \mathbf{m}_{i1}^T \\ \mathbf{m}_{i2}^T \end{bmatrix} \\ \mathbf{m}_{i1}^T \mathbf{C} \mathbf{C}^T \mathbf{m}_{i2} &= 0 \end{aligned}$$

- Two options:
 - Solve for $\mathbf{C}$ (Newton's method, quadratic)
 - Solve linearly $\mathbf{L} = \mathbf{C} \mathbf{C}^T$
 - Recover $\mathbf{C}$ from $\mathbf{L}$ by SVD or Cholesky decomposition: $\mathbf{L} = \mathbf{C} \mathbf{C}^T$
- Update $\mathbf{M}$ and $\mathbf{S}$: $\mathbf{M}' = \mathbf{M} \mathbf{C}$, $\mathbf{S}' = \mathbf{C}^{-1} \mathbf{S}$

2/2/2006

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Algorithm summary



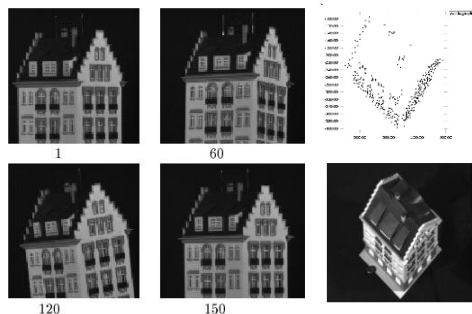
- Given: m images and n features $\mathbf{x}_{ij}$
- For each image i , center the feature coordinates
- Construct a $2m \times n$ measurement matrix $\mathbf{D}$:
 - Column j contains the projection of point j in all views
 - Row i contains one coordinate of the projections of all the n points in image i
- Factorize $\mathbf{D}$:
 - Compute SVD: $\mathbf{D} = \mathbf{U} \mathbf{W} \mathbf{V}^T$
 - Create $\mathbf{U}_3$ by taking the first 3 columns of $\mathbf{U}$
 - Create $\mathbf{V}_3$ by taking the first 3 columns of $\mathbf{V}$
 - Create $\mathbf{W}_3$ by taking the upper left 3×3 block of $\mathbf{W}$
- Create the motion and shape matrices:
 - $\mathbf{M} = \mathbf{U}_3 \mathbf{W}_3^{1/2}$ and $\mathbf{S} = \mathbf{W}_3^{1/2} \mathbf{V}_3^T$ (or $\mathbf{M} = \mathbf{U}_3$ and $\mathbf{S} = \mathbf{W}_3 \mathbf{V}_3^T$)
- Eliminate affine ambiguity

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Reconstruction results



C. Tomasi and T. Kanade. Shape and motion from image streams under orthography: A factorization method. *IJCV*, 9(2):137-154, November 1992.

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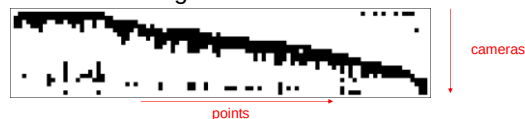
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Dealing with missing data



- So far, we have assumed that all points are visible in all views
- In reality, the measurement matrix typically looks something like this:



2/2/2006

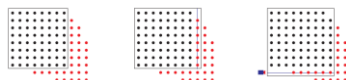
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Dealing with missing data



- Possible solution: decompose matrix into dense sub-blocks, factorize each sub-block, and fuse the results
 - Finding dense maximal sub-blocks of the matrix is NP-complete (equivalent to finding maximal cliques in a graph)
- Incremental bilinear refinement



- (1) Perform factorization on a dense sub-block
- (2) Solve for a new 3D point visible by at least two known cameras (linear least squares)
- (3) Solve for a new camera that sees at least three known 3D points (linear least squares)

F. Rothganger, S. Lazebnik, C. Schmid, and J. Ponce. Segmenting, Modeling, and Matching Video Clips Containing Multiple Moving Objects. *PAMI* 2007.

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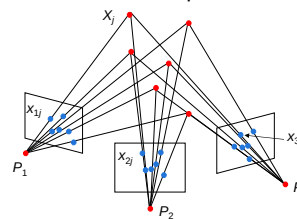
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Projective structure from motion



- Given: m images of n fixed 3D points

$$\mathbf{x}_{ij} = \mathbf{P}_i \mathbf{X}_j, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$
- Problem: estimate m projection matrices $\mathbf{P}_i$ and n 3D points $\mathbf{X}_j$ from the mn correspondences $\mathbf{x}_{ij}$



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Projective structure from motion



- Given: m images of n fixed 3D points

$$\mathbf{x}_{ij} = \mathbf{P}_i \mathbf{X}_j, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

- Problem: estimate m projection matrices $\mathbf{P}_i$ and n 3D points $\mathbf{X}_j$ from the mn correspondences $\mathbf{x}_{ij}$
- With no calibration info, cameras and points can only be recovered up to a 4x4 projective transformation $\mathbf{Q}$:

$$\mathbf{X} \rightarrow \mathbf{Q}\mathbf{X}, \quad \mathbf{P} \rightarrow \mathbf{P}\mathbf{Q}^{-1}$$

- We can solve for structure and motion when $2mn \geq 11m + 3n - 15$
- For two cameras, at least 7 points are needed

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Projective SFM: Two-camera case



- Compute fundamental matrix $\mathbf{F}$ between the two views
- First camera matrix: $[\mathbf{I}|\mathbf{0}]$
- Second camera matrix: $[\mathbf{A}|\mathbf{b}]$
- Then $\mathbf{b}$ is the epipole ($\mathbf{F}^T \mathbf{b} = \mathbf{0}$), $\mathbf{A} = -[\mathbf{b}_\times] \mathbf{F}$

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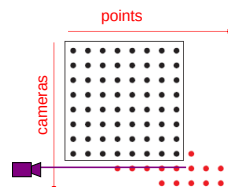
Sequential structure from motion



- Initialize motion from two images using fundamental matrix

- Initialize structure by triangulation

- For each additional view:
 - Determine projection matrix of new camera using all the known 3D points that are visible in its image – *calibration*



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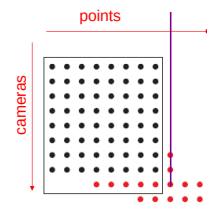
Sequential structure from motion



- Initialize motion from two images using fundamental matrix

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- For each additional view:
 - Determine projection matrix of new camera using all the known 3D points that are visible in its image – *calibration*
 - Refine and extend structure: compute new 3D points, re-optimize existing points that are also seen by this camera – *triangulation*



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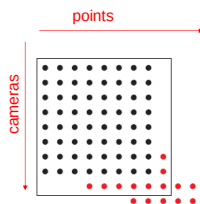
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 - Determine projection matrix of new camera using all the known 3D points that are visible in its image – *calibration*
 - Refine and extend structure: compute new 3D points, re-optimize existing points that are also seen by this camera – *triangulation*



- Refine structure and motion: bundle adjustment

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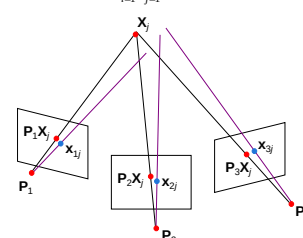
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Bundle adjustment



- Non-linear method for refining structure and motion
- Minimizing reprojection error

$$E(\mathbf{P}, \mathbf{X}) = \sum_{i=1}^m \sum_{j=1}^n w_{ij} D(\mathbf{x}_{ij}, \mathbf{P}_i \mathbf{X}_j)^2$$



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Bundle adjustment



- Assume n 3D points are seen in m views. Illustration with $n = 7, m = 3$
- Let $\mathbf{x}_{ij}$ be the projection of the i -th point on image j , $\mathbf{a}_j$ the vector of parameters for camera j and $\mathbf{b}_i$ the vector of parameters for point i
- BA minimizes the *reprojection error* over all point and camera parameters: $\min_{\mathbf{a}_j, \mathbf{b}_i} \sum_{i=1}^n \sum_{j=1}^m v_{ij} d(\mathbf{Q}(\mathbf{a}_j, \mathbf{b}_i), \mathbf{x}_{ij})^2$,
 $\mathbf{Q}(\mathbf{a}_j, \mathbf{b}_i)$ being the predicted projection of point i on image j , $d(\cdot, \cdot)$ the Euclidean distance between image points and $v_{ij} = 1$ iff point i is visible in image j
- This is a large problem: if n_i, λ are the dimensions of the $\mathbf{a}_j$ & $\mathbf{b}_i$, the total number of parameters involved in BA is $mK + n\lambda$

Bundle adjustment

- A parameter vector $\mathbf{P}$ is defined by partitioning parameters as $\mathbf{P} = (\mathbf{a}_1^T, \dots, \mathbf{a}_m^T, \mathbf{b}_1^T, \dots, \mathbf{b}_n^T)^T$
- A measurement vector $\mathbf{X}$ is defined as $(\mathbf{x}_{11}^T, \dots, \mathbf{x}_{1m}^T, \mathbf{x}_{21}^T, \dots, \mathbf{x}_{2m}^T, \dots, \mathbf{x}_{n1}^T, \dots, \mathbf{x}_{nm}^T)^T$
- For each parameter vector, an estimated measurement $\hat{\mathbf{X}}$ is $(\hat{\mathbf{x}}_{11}^T, \dots, \hat{\mathbf{x}}_{1m}^T, \hat{\mathbf{x}}_{21}^T, \dots, \hat{\mathbf{x}}_{2m}^T, \dots, \hat{\mathbf{x}}_{n1}^T, \dots, \hat{\mathbf{x}}_{nm}^T)^T$ and the corresponding error $(\epsilon_{11}^T, \dots, \epsilon_{1m}^T, \epsilon_{21}^T, \dots, \epsilon_{2m}^T, \dots, \epsilon_{n1}^T, \dots, \epsilon_{nm}^T)^T$, where $\hat{\mathbf{x}}_{ij} \equiv \mathbf{Q}(\mathbf{a}_j, \mathbf{b}_i)$ and $\epsilon_{ij} \equiv \mathbf{x}_{ij} - \hat{\mathbf{x}}_{ij} \forall i, j$
- With the above definitions, BA corresponds to minimizing $\sum_{i=1}^n \sum_{j=1}^m \|\epsilon_{ij}\|^2 = \|\mathbf{X} - \hat{\mathbf{X}}\|^2$ over $\mathbf{P}$, which is a nonlinear least squares problem

Levenberg-Marquardt

- Let $f: \mathcal{R}^m \rightarrow \mathcal{R}^n$. Given an initial estimate $\mathbf{p}_0 \in \mathcal{R}^m$ and a measurement vector $\mathbf{x} \in \mathcal{R}^n$, LM seeks to find $\mathbf{p}^+$ minimizing $\epsilon^T \epsilon$, $\epsilon = \mathbf{x} - f(\mathbf{p})$
- Note that this is a (nonlinear) least squares problem since $\epsilon^T \epsilon = \|\mathbf{x} - f(\mathbf{p})\|^2$, $\|\cdot\|$ being the L2 norm
- The minimizer can be found by the Gauss-Newton method, which iteratively linearizes f at $\mathbf{p}$ and determines incremental update steps δ_p by solving the *normal equations* $\mathbf{J}^T \mathbf{J} \delta_p = \mathbf{J}^T \epsilon$, $\mathbf{J}$ being the Jacobian of f at $\mathbf{p}$ and $\mathbf{J}^T \mathbf{J}$ the approximate Hessian of $\|\epsilon\|^2$
- To ensure convergence, LM uses *damping*, i.e. adaptively alters the diagonal elements of $\mathbf{J}^T \mathbf{J}$ and solves the *augmented normal equations* $(\mathbf{J}^T \mathbf{J} + \mu \mathbf{I}) \delta_p = \mathbf{J}^T \epsilon$, $\mu > 0$

Levenberg-Marquardt

- The Jacobian $\mathbf{J} = \frac{\partial \hat{\mathbf{X}}}{\partial \mathbf{P}}$ has a block structure $[\mathbf{A} | \mathbf{B}]$, where $\mathbf{A} = \left[\frac{\partial \hat{\mathbf{X}}}{\partial \mathbf{a}} \right]$ and $\mathbf{B} = \left[\frac{\partial \hat{\mathbf{X}}}{\partial \mathbf{b}} \right]$
- The LM updating vector δ is partitioned as $(\delta_a^T, \delta_b^T)^T$
- The normal equations become
$$\begin{bmatrix} \mathbf{A}^T \mathbf{A} & \mathbf{A}^T \mathbf{B} \\ \mathbf{B}^T \mathbf{A} & \mathbf{B}^T \mathbf{B} \end{bmatrix} \begin{pmatrix} \delta_a \\ \delta_b \end{pmatrix} = \begin{pmatrix} \mathbf{A}^T \epsilon \\ \mathbf{B}^T \epsilon \end{pmatrix}$$
- The lhs matrix above is sparse due to $\mathbf{A}$ and $\mathbf{B}$ being sparse: $\frac{\partial \hat{\mathbf{x}}_{ij}}{\partial \mathbf{a}_k} = 0, \forall j \neq k$ and $\frac{\partial \hat{\mathbf{x}}_{ij}}{\partial \mathbf{b}_k} = 0, \forall i \neq k$
- This is the so-called *primary structure* of BA

Example (n=4,m=3)

- Assume all points are seen in all images
- The measurement vector $\mathbf{x} = (\mathbf{x}_{11}^T, \mathbf{x}_{12}^T, \mathbf{x}_{13}^T, \mathbf{x}_{21}^T, \mathbf{x}_{22}^T, \mathbf{x}_{23}^T, \mathbf{x}_{31}^T, \mathbf{x}_{32}^T, \mathbf{x}_{33}^T, \mathbf{x}_{41}^T, \mathbf{x}_{42}^T, \mathbf{x}_{43}^T)^T$
- The parameter vector $\mathbf{P} = (\mathbf{a}_1^T, \mathbf{a}_2^T, \mathbf{a}_3^T, \mathbf{b}_1^T, \mathbf{b}_2^T, \mathbf{b}_3^T, \mathbf{b}_4^T)^T$
- The LM updating vector $\delta = (\delta_{a_1}^T, \delta_{a_2}^T, \delta_{a_3}^T, \delta_{b_1}^T, \delta_{b_2}^T, \delta_{b_3}^T, \delta_{b_4}^T)^T$
- Let $\mathbf{A}_{ij} = \frac{\partial \hat{\mathbf{x}}_{ij}}{\partial \mathbf{a}_j}$ and $\mathbf{B}_{ij} = \frac{\partial \hat{\mathbf{x}}_{ij}}{\partial \mathbf{b}_i}$

Jacobian (n=4,m=3)

$$\frac{\partial \hat{\mathbf{X}}}{\partial \mathbf{P}} = \begin{matrix} & \mathbf{a}_1^T & \mathbf{a}_2^T & \mathbf{a}_3^T & \mathbf{b}_1^T & \mathbf{b}_2^T & \mathbf{b}_3^T & \mathbf{b}_4^T \\ \begin{matrix} \mathbf{x}_{11} \\ \mathbf{x}_{12} \\ \mathbf{x}_{13} \\ \mathbf{x}_{21} \\ \mathbf{x}_{22} \\ \mathbf{x}_{23} \\ \mathbf{x}_{31} \\ \mathbf{x}_{32} \\ \mathbf{x}_{33} \\ \mathbf{x}_{41} \\ \mathbf{x}_{42} \\ \mathbf{x}_{43} \end{matrix} & \begin{pmatrix} \mathbf{A}_{11} & 0 & 0 & \mathbf{B}_{11} & 0 & 0 & 0 \\ 0 & \mathbf{A}_{12} & 0 & \mathbf{B}_{12} & 0 & 0 & 0 \\ 0 & 0 & \mathbf{A}_{13} & \mathbf{B}_{13} & 0 & 0 & 0 \\ \mathbf{A}_{21} & 0 & 0 & 0 & \mathbf{B}_{21} & 0 & 0 \\ 0 & \mathbf{A}_{22} & 0 & 0 & \mathbf{B}_{22} & 0 & 0 \\ 0 & 0 & \mathbf{A}_{23} & 0 & \mathbf{B}_{23} & 0 & 0 \\ \mathbf{A}_{31} & 0 & 0 & 0 & 0 & \mathbf{B}_{31} & 0 \\ 0 & \mathbf{A}_{32} & 0 & 0 & 0 & \mathbf{B}_{32} & 0 \\ 0 & 0 & \mathbf{A}_{33} & 0 & 0 & \mathbf{B}_{33} & 0 \\ \mathbf{A}_{41} & 0 & 0 & 0 & 0 & 0 & \mathbf{B}_{41} \\ 0 & \mathbf{A}_{42} & 0 & 0 & 0 & 0 & \mathbf{B}_{42} \\ 0 & 0 & \mathbf{A}_{43} & 0 & 0 & 0 & \mathbf{B}_{43} \end{pmatrix} \end{matrix}$$

Hessian (n=4,m=3)

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- Approximate Hessian in block form:

$$J^T J = \begin{pmatrix} \mathbf{a}_1^T & \mathbf{a}_2^T & \mathbf{a}_3^T & \mathbf{b}_1^T & \mathbf{b}_2^T & \mathbf{b}_3^T & \mathbf{b}_4^T \\ \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 & \mathbf{b}_4 \\ \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 & \mathbf{b}_2 & \mathbf{b}_3 & \mathbf{b}_4 & \mathbf{b}_5 \\ \mathbf{a}_3 & \mathbf{a}_4 & \mathbf{a}_5 & \mathbf{b}_3 & \mathbf{b}_4 & \mathbf{b}_5 & \mathbf{b}_6 \\ \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 & \mathbf{b}_4 & \mathbf{b}_5 & \mathbf{b}_6 & \mathbf{b}_7 \\ \mathbf{b}_2 & \mathbf{b}_3 & \mathbf{b}_4 & \mathbf{b}_5 & \mathbf{b}_6 & \mathbf{b}_7 & \mathbf{b}_8 \\ \mathbf{b}_3 & \mathbf{b}_4 & \mathbf{b}_5 & \mathbf{b}_6 & \mathbf{b}_7 & \mathbf{b}_8 & \mathbf{b}_9 \\ \mathbf{b}_4 & \mathbf{b}_5 & \mathbf{b}_6 & \mathbf{b}_7 & \mathbf{b}_8 & \mathbf{b}_9 & \mathbf{b}_{10} \end{pmatrix} \equiv \begin{pmatrix} \mathbf{U} & \mathbf{W} \\ \mathbf{W}^T & \mathbf{V} \end{pmatrix},$$

(2)

where

$$\mathbf{U}_j \equiv \sum_{i=1}^4 \mathbf{A}_{ij}^T \mathbf{A}_{ij},$$

$$\mathbf{V}_i \equiv \sum_{j=1}^3 \mathbf{B}_{ij}^T \mathbf{B}_{ij},$$

$$\mathbf{W}_{ij} = \mathbf{A}_{ij}^T \mathbf{B}_{ij}$$

- The above generalize directly to arbitrary n and m
- $\mathbf{U}$ and $\mathbf{V}$ are block diagonal, $\mathbf{W}$ arbitrarily sparse

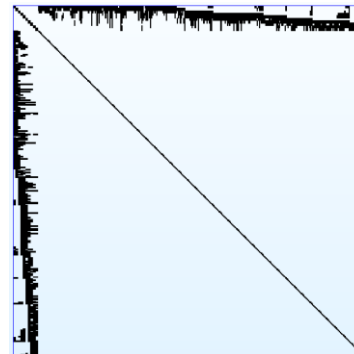
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Hessian (real problem)

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Black: non-zero

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Bundle adjustment

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- Can be solved efficiently due to sparseness
- Good news: open source libraries exist, e.g., sba : A Generic Sparse Bundle Adjustment C/C++ Package (<http://users.ics.forth.gr/~lourakis/sba/>)

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Self-calibration

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- Self-calibration (auto-calibration) is the process of determining intrinsic camera parameters directly from uncalibrated images
- For example, when the images are acquired by a single moving camera, we can use the constraint that the intrinsic parameter matrix remains fixed for all the images
 - Compute initial projective reconstruction and find 3D projective transformation matrix $\mathbf{Q}$ such that all camera matrices are in the form $\mathbf{P}_i = \mathbf{K} [\mathbf{R}_i | \mathbf{t}_i]$
- Can use constraints on the form of the calibration matrix: zero skew

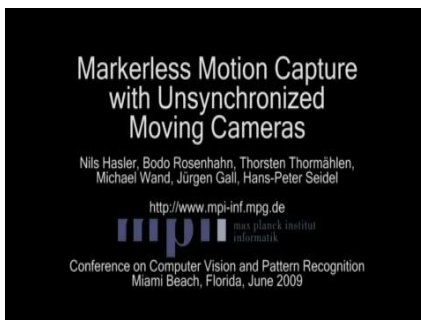
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Example

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N. Hasler et al. Markerless Motion Capture with Unsynchronized Moving Cameras. CVPR'09.

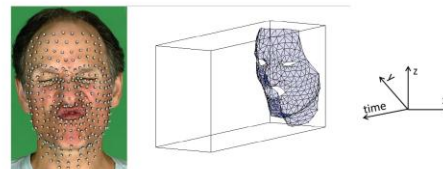
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NONRIGID STRUCTURE

3D Structure That Deforms Over Time



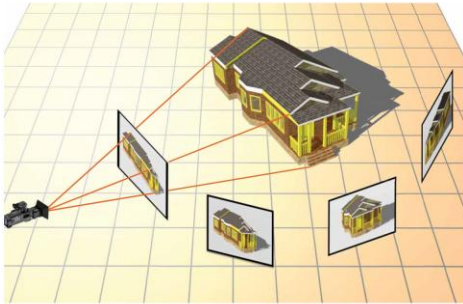
4D DYNAMIC STRUCTURE

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RIGID STRUCTURE FROM MOTION

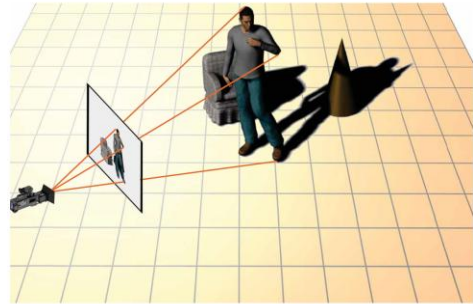


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NONRIGID STRUCTURE FROM MOTION

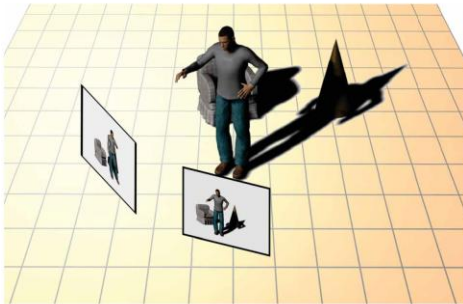


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NONRIGID STRUCTURE FROM MOTION

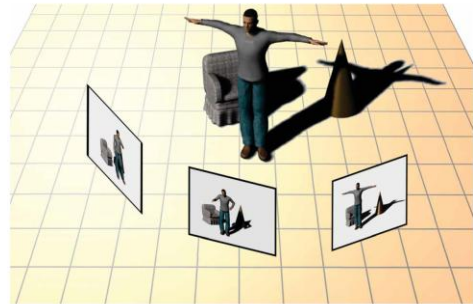


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NONRIGID STRUCTURE FROM MOTION



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NONRIGID STRUCTURE FROM MOTION



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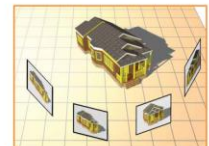
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FACTORIZATION METHOD FOR RIGID SFM

Kontsevich *et al.* 1987, Tomasi and Kanade 1992

ASSUMPTIONS

- Orthographic Camera
- At least 3 images
- Rigid Scene
- Camera Motion
- Corresponding points available



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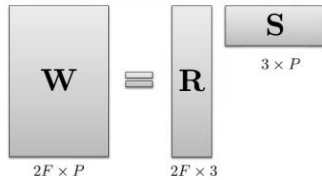
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FACTORIZATION METHOD FOR RIGID SFM

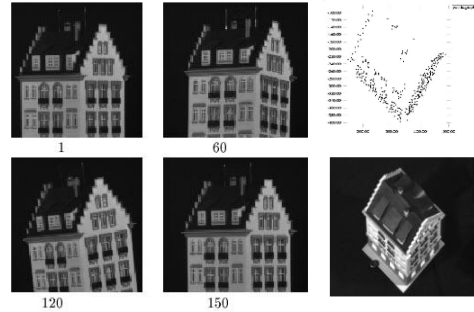
Kontsevich *et al.* 1987, Tomasi and Kanade 1992

PROJECTION OF P 3D POINTS IN F IMAGES



$$\mathbf{W}_{\text{measurement}} = \mathbf{R}_{\text{motion}} \times \mathbf{S}_{\text{shape}}$$

Recall: Reconstruction results



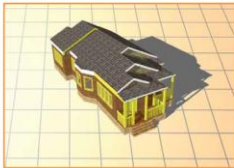
C. Tomasi and T. Kanade, Shape and motion from image streams under orthography: A factorization method, *IJCV*, 9(2):137-154, November 1992.

NONRIGID STRUCTURE

3D Structure That Deforms Over Time

RIGID STRUCTURE

$$\mathbf{S}_{3 \times P} = \begin{bmatrix} X_1 & X_2 & \dots & X_P \\ Y_1 & Y_2 & \dots & Y_P \\ Z_1 & Z_2 & \dots & Z_P \end{bmatrix}$$

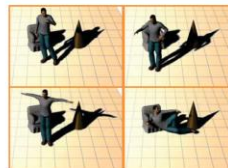


NONRIGID STRUCTURE

3D Structure That Deforms Over Time

RIGID STRUCTURE

$$\mathbf{S}_{3 \times P} = \begin{bmatrix} X_1 & X_2 & \dots & X_P \\ Y_1 & Y_2 & \dots & Y_P \\ Z_1 & Z_2 & \dots & Z_P \end{bmatrix}$$



NONRIGID STRUCTURE

$$\mathbf{S}_{3F \times P} = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1P} \\ Y_{11} & Y_{12} & \dots & Y_{1P} \\ Z_{11} & Z_{12} & \dots & Z_{1P} \\ \vdots & \vdots & \ddots & \vdots \\ X_{F1} & X_{F2} & \dots & X_{FP} \\ Y_{F1} & Y_{F2} & \dots & Y_{FP} \\ Z_{F1} & Z_{F2} & \dots & Z_{FP} \end{bmatrix}$$

NONRIGID STRUCTURE

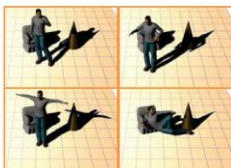
3D Structure That Deforms Over Time

RIGID STRUCTURE

$$\mathbf{S}_{3 \times P} = \begin{bmatrix} X_1 & X_2 & \dots & X_P \\ Y_1 & Y_2 & \dots & Y_P \\ Z_1 & Z_2 & \dots & Z_P \end{bmatrix}$$

NONRIGID STRUCTURE

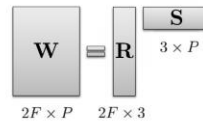
$$\mathbf{S}_{3F \times P} = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1P} \\ X_{21} & X_{22} & \dots & X_{2P} \\ \vdots & \vdots & \ddots & \vdots \\ X_{F1} & X_{F2} & \dots & X_{FP} \end{bmatrix}$$



NONRIGID STRUCTURE FROM MOTION

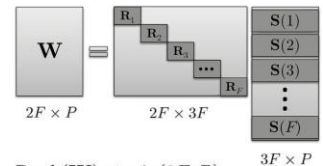
Comparison with Rigid Structure from Motion

RIGID SFM



$$\text{Rank}(\mathbf{W}) \leq 3$$

NONRIGID SFM



$$\text{Rank}(\mathbf{W}) \leq \min(2F, P)$$

NONRIGID STRUCTURE FROM MOTION

Explosion of Unknowns

Example: Given a 40 second video with 100 tracked points

RIGID SFM

- Inputs:
100 pts x 40 sec x 30 fps x 2 (x, y)
= 240,000 observations
- Unknowns:
100 points x 3 (X, Y, Z)
= **300** unknowns

NONRIGID SFM

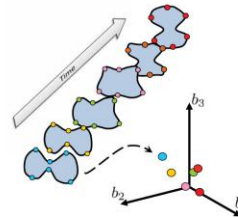
- Inputs:
100 pts x 40 sec x 30 fps x 2
= 240,000 observations
- Unknowns:
100 points x 40 sec x 30 fps x 3
= **360,000** unknowns

NONRIGID STRUCTURE FROM MOTION

Two Major Approaches

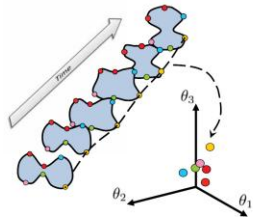
Shape Basis

3D points at each time instant lie in a low dimensional subspace



Trajectory Basis

Trajectory of each point over time lies in a low dimensional subspace



EXAMPLES OF APPLICATIONS

Motion-Capture Cleanup



Video



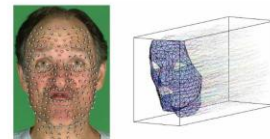
Unlabeled Data
Input



Reconstruction
Output

Disney Research, Pittsburgh

DYNAMIC STRUCTURE



$$S_{3F \times P} = \begin{bmatrix} X_{11} & X_{12} & \cdots & X_{1P} \\ X_{21} & X_{22} & \cdots & X_{2P} \\ \vdots & \vdots & \ddots & \vdots \\ X_{F1} & X_{F2} & \cdots & X_{FP} \end{bmatrix} \begin{matrix} \text{Trajectory} \\ \downarrow \end{matrix}$$

Shape

DYNAMIC STRUCTURE UNDER ORTHOGRAPHIC PROJECTION

$$\begin{bmatrix} X_{11} & \cdots & X_{1P} \\ X_{21} & \cdots & X_{2P} \\ \vdots & \ddots & \vdots \\ X_{F1} & \cdots & X_{FP} \end{bmatrix} = \begin{bmatrix} R_1 & & \\ & R_2 & \\ & & \ddots \\ & & & R_F \end{bmatrix} \begin{bmatrix} X_{11} & \cdots & X_{1P} \\ X_{21} & \cdots & X_{2P} \\ \vdots & \ddots & \vdots \\ X_{F1} & \cdots & X_{FP} \end{bmatrix}$$

$$W = RX$$

LINEAR SHAPE MODEL

[T. Cootes et al. 91, Bregler et al. 97]



$$\begin{bmatrix} X_{11} & \cdots & X_{1P} \\ X_{21} & \cdots & X_{2P} \\ \vdots & \ddots & \vdots \\ X_{F1} & \cdots & X_{FP} \end{bmatrix}$$

$$= \alpha_1 \times \text{[Face 1]} + \alpha_2 \times \text{[Face 2]} + \cdots + \alpha_k \times \text{[Face k]}$$

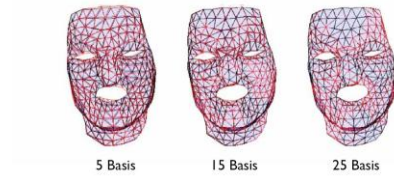
LINEAR SHAPE MODEL

$$\begin{bmatrix} \mathbf{X}_{11} & \cdots & \mathbf{X}_{1P} \\ \mathbf{X}_{21} & & \mathbf{X}_{2P} \\ \vdots & & \vdots \\ \mathbf{X}_{F1} & \cdots & \mathbf{X}_{FP} \end{bmatrix} = \begin{bmatrix} \omega_{11} & \cdots & \omega_{1k} \\ \omega_{21} & & \omega_{2k} \\ \vdots & & \vdots \\ \omega_{F1} & \cdots & \omega_{Fk} \end{bmatrix} \begin{bmatrix} -\mathbf{b}_1- \\ -\mathbf{b}_2- \\ \vdots \\ -\mathbf{b}_k- \end{bmatrix}$$

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LINEAR SHAPE MODEL
RECONSTRUCTION

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LINEAR SHAPE MODEL
UNDER ORTHOGRAPHIC PROJECTION

$$\begin{bmatrix} \mathbf{X}_{11} & \cdots & \mathbf{X}_{1P} \\ \mathbf{X}_{21} & & \mathbf{X}_{2P} \\ \vdots & & \vdots \\ \mathbf{X}_{F1} & \cdots & \mathbf{X}_{FP} \end{bmatrix} = \begin{bmatrix} \mathbf{R}_1 & & \\ & \ddots & \\ & & \mathbf{R}_F \end{bmatrix} \begin{bmatrix} \mathbf{X}_{11} & \cdots & \mathbf{X}_{1P} \\ \mathbf{X}_{21} & & \mathbf{X}_{2P} \\ \vdots & & \vdots \\ \mathbf{X}_{F1} & \cdots & \mathbf{X}_{FP} \end{bmatrix}$$

$2F \times P$ $2F \times 3F (6F)$ $3F \times P$

$$= \begin{bmatrix} \mathbf{R}_1 & & \\ & \ddots & \\ & & \mathbf{R}_F \end{bmatrix} \begin{bmatrix} \omega_{11} & \cdots & \omega_{1k} \\ \omega_{21} & & \omega_{2k} \\ \vdots & & \vdots \\ \omega_{F1} & \cdots & \omega_{Fk} \end{bmatrix} \begin{bmatrix} -\mathbf{b}_1- \\ -\mathbf{b}_2- \\ \vdots \\ -\mathbf{b}_k- \end{bmatrix}$$

$2F \times 3F (6F)$ $3F \times 3k$ $3k \times P$

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CHALLENGE
TRILINEAR ESTIMATION

$$\begin{bmatrix} \mathbf{R}_1 & & \\ & \ddots & \\ & & \mathbf{R}_F \end{bmatrix} \begin{bmatrix} \omega_{11} & \cdots & \omega_{1k} \\ \omega_{21} & & \omega_{2k} \\ \vdots & & \vdots \\ \omega_{F1} & \cdots & \omega_{Fk} \end{bmatrix} \begin{bmatrix} -\mathbf{b}_1- \\ -\mathbf{b}_2- \\ \vdots \\ -\mathbf{b}_k- \end{bmatrix}$$

$\mathbf{W} = \mathbf{R}\mathbf{\Omega}\mathbf{B}$

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BREGLER et al. 2000
Nested SVD

$$\begin{bmatrix} \mathbf{X}_{11} & \cdots & \mathbf{X}_{1P} \\ \mathbf{X}_{21} & & \mathbf{X}_{2P} \\ \vdots & & \vdots \\ \mathbf{X}_{F1} & \cdots & \mathbf{X}_{FP} \end{bmatrix} = \begin{bmatrix} \mathbf{R}_1 & & \\ & \ddots & \\ & & \mathbf{R}_F \end{bmatrix} \begin{bmatrix} \omega_{11} & \cdots & \omega_{1k} \\ \omega_{21} & & \omega_{2k} \\ \vdots & & \vdots \\ \omega_{F1} & \cdots & \omega_{Fk} \end{bmatrix} \begin{bmatrix} -\mathbf{b}_1- \\ -\mathbf{b}_2- \\ \vdots \\ -\mathbf{b}_k- \end{bmatrix}$$

$$= \begin{bmatrix} \omega_{11}\mathbf{R}_1 & \cdots & \omega_{1k}\mathbf{R}_1 \\ \omega_{21}\mathbf{R}_2 & & \omega_{2k}\mathbf{R}_2 \\ \vdots & & \vdots \\ \omega_{F1}\mathbf{R}_F & \cdots & \omega_{Fk}\mathbf{R}_F \end{bmatrix} \begin{bmatrix} -\mathbf{b}_1- \\ -\mathbf{b}_2- \\ \vdots \\ -\mathbf{b}_k- \end{bmatrix}$$

$2F \times 3k$ $3k \times P$

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BREGLER et al. 2000
Outer SVD

$$\mathbf{W} = \begin{bmatrix} \mathbf{X}_{11} & \cdots & \mathbf{X}_{1P} \\ \mathbf{X}_{21} & & \mathbf{X}_{2P} \\ \vdots & & \vdots \\ \mathbf{X}_{F1} & \cdots & \mathbf{X}_{FP} \end{bmatrix} = \begin{bmatrix} \omega_{11}\mathbf{R}_1 & \cdots & \omega_{1k}\mathbf{R}_1 \\ \omega_{21}\mathbf{R}_2 & & \omega_{2k}\mathbf{R}_2 \\ \vdots & & \vdots \\ \omega_{F1}\mathbf{R}_F & \cdots & \omega_{Fk}\mathbf{R}_F \end{bmatrix} \begin{bmatrix} -\mathbf{b}_1- \\ -\mathbf{b}_2- \\ \vdots \\ -\mathbf{b}_k- \end{bmatrix}$$

$2F \times 3k$ $3k \times P$

SVD

$$\mathbf{W} = \mathbf{U}\mathbf{D}\mathbf{V}^T$$

$$\mathbf{W} = (\mathbf{U}\mathbf{D}^{\frac{1}{2}})(\mathbf{D}^{\frac{1}{2}}\mathbf{V}^T)$$

$$\mathbf{W} = \hat{\mathbf{U}}\hat{\mathbf{B}}$$

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BREGLER et al. 2000

Inner SVD

$$\mathbf{W} = \hat{\mathbf{H}}\mathbf{B}$$

$$\mathbf{H} = \begin{bmatrix} \omega_{11}\mathbf{R}_1 & \cdots & \omega_{1k}\mathbf{R}_1 \\ \omega_{21}\mathbf{R}_2 & & \omega_{2k}\mathbf{R}_2 \\ \vdots & & \vdots \\ \omega_{F1}\mathbf{R}_F & \cdots & \omega_{Fk}\mathbf{R}_F \end{bmatrix}$$

$$\mathbf{h}_1 = \begin{bmatrix} \omega_{11}r_1^1 & \omega_{11}r_1^2 & \omega_{11}r_1^3 & \cdots & \omega_{1k}r_1^1 & \omega_{1k}r_1^2 & \omega_{1k}r_1^3 \\ \omega_{11}r_2^1 & \omega_{11}r_2^2 & \omega_{11}r_2^3 & \cdots & \omega_{1k}r_2^1 & \omega_{1k}r_2^2 & \omega_{1k}r_2^3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ \omega_{F1}r_F^1 & \omega_{F1}r_F^2 & \omega_{F1}r_F^3 & \cdots & \omega_{Fk}r_F^1 & \omega_{Fk}r_F^2 & \omega_{Fk}r_F^3 \end{bmatrix}$$

$$\mathbf{h}'_1 = \begin{bmatrix} \omega_{11}r_1^1 & \omega_{11}r_1^2 & \omega_{11}r_1^3 & \omega_{11}r_1^4 & \omega_{11}r_1^5 & \omega_{11}r_1^6 \\ \omega_{11}r_2^1 & \omega_{11}r_2^2 & \omega_{11}r_2^3 & \omega_{11}r_2^4 & \omega_{11}r_2^5 & \omega_{11}r_2^6 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \omega_{F1}r_F^1 & \omega_{F1}r_F^2 & \omega_{F1}r_F^3 & \omega_{F1}r_F^4 & \omega_{F1}r_F^5 & \omega_{F1}r_F^6 \end{bmatrix} = \begin{bmatrix} \omega_{11} \\ \vdots \\ \omega_{12} \end{bmatrix} \begin{bmatrix} r_1^1 & r_1^2 & r_1^3 & r_1^4 & r_1^5 & r_1^6 \end{bmatrix}$$

rank 1

SVD $\mathbf{h}'_1 = \mathbf{u}\mathbf{d}\mathbf{v}^T = \hat{\mathbf{W}}\hat{\mathbf{r}}$

METRIC RECTIFICATION USING ORTHONORMALITY CONSTRAINTS

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BREGLER et al. 2000

OVERVIEW

- **OUTER SVD:** PERFORM SVD ON $\mathbf{W}$ TO GET ESTIMATES OF:
 - $\mathbf{H}$: CAMERA PROJECTIONS AND COEFFICIENTS
- **INNER SVD:** PERFORM SVD ON $\mathbf{H}$ TO GET ESTIMATES OF:
 - $\mathbf{\Omega}$: COEFFICIENTS
 - $\mathbf{R}$: CAMERA PROJECTIONS
- **METRIC RECTIFY USING ORTHONORMALITY CONSTRAINTS**
- $\mathbf{B}$: THE SHAPE BASIS

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RESULTS




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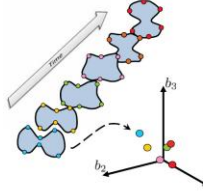
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NONRIGID STRUCTURE FROM MOTION

Two Major Approaches

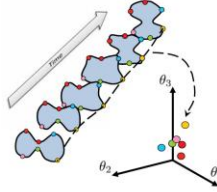
Shape Basis

3D points at each time instant lie in a low dimensional subspace



Trajectory Basis

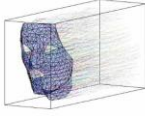
Trajectory of each point over time lies in a low dimensional subspace



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DYNAMIC STRUCTURE

$$\mathbf{S}_{3F \times P} = \begin{bmatrix} \mathbf{X}_{11} & \mathbf{X}_{12} & \cdots & \mathbf{X}_{1P} \\ \mathbf{X}_{21} & \mathbf{X}_{22} & \cdots & \mathbf{X}_{2P} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{X}_{F1} & \mathbf{X}_{F2} & \cdots & \mathbf{X}_{FP} \end{bmatrix}$$

Shape

Trajectory

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DYNAMIC STRUCTURE

Shape Representation

$$\mathbf{S}_{3F \times P} = \begin{bmatrix} \mathbf{X}_{11} & \mathbf{X}_{12} & \cdots & \mathbf{X}_{1P} \\ \mathbf{X}_{21} & \mathbf{X}_{22} & \cdots & \mathbf{X}_{2P} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{X}_{F1} & \mathbf{X}_{F2} & \cdots & \mathbf{X}_{FP} \end{bmatrix}$$

Shape

LINEAR SHAPE MODEL

$$\text{Face} = \omega_1 \times \text{Face}_1 + \omega_2 \times \text{Face}_2 + \dots + \omega_k \times \text{Face}_k$$

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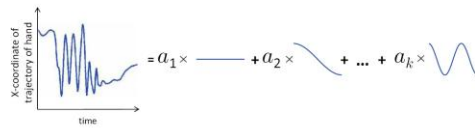
DYNAMIC STRUCTURE

Trajectory Representation



$$S_{3F \times P} = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1P} \\ X_{21} & X_{22} & \dots & X_{2P} \\ \vdots & \vdots & \ddots & \vdots \\ X_{F1} & X_{F2} & \dots & X_{FP} \end{bmatrix} \quad \text{Trajectory}$$

LINEAR TRAJECTORY MODEL



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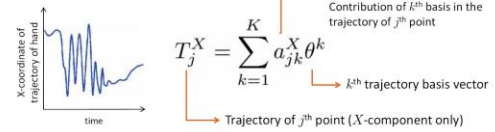
DYNAMIC STRUCTURE

Trajectory Representation



$$S_{3F \times P} = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1P} \\ X_{21} & X_{22} & \dots & X_{2P} \\ \vdots & \vdots & \ddots & \vdots \\ X_{F1} & X_{F2} & \dots & X_{FP} \end{bmatrix} \quad \text{Trajectory}$$

LINEAR TRAJECTORY MODEL



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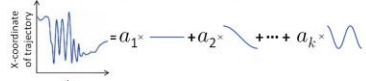
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TRAJECTORY REPRESENTATION OF DYNAMIC STRUCTURE

$$T_j^X = \sum_{k=1}^K a_{jk}^X \theta^k \quad T_j^Y = \sum_{k=1}^K a_{jk}^Y \theta^k \quad T_j^Z = \sum_{k=1}^K a_{jk}^Z \theta^k$$

$$\begin{bmatrix} X_{1j} \\ X_{2j} \\ \vdots \\ X_{Fj} \end{bmatrix} = a_{j1}^X \begin{bmatrix} \theta_1^1 \\ \theta_2^1 \\ \vdots \\ \theta_F^1 \end{bmatrix} + a_{j2}^X \begin{bmatrix} \theta_1^2 \\ \theta_2^2 \\ \vdots \\ \theta_F^2 \end{bmatrix} + \dots + a_{jK}^X \begin{bmatrix} \theta_1^K \\ \theta_2^K \\ \vdots \\ \theta_F^K \end{bmatrix}$$



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TRAJECTORY REPRESENTATION OF DYNAMIC STRUCTURE

$$\begin{bmatrix} X_{1j} \\ X_{2j} \\ \vdots \\ X_{Fj} \end{bmatrix} = a_{j1}^X \begin{bmatrix} \theta_1^1 \\ \theta_2^1 \\ \vdots \\ \theta_F^1 \end{bmatrix} + a_{j2}^X \begin{bmatrix} \theta_1^2 \\ \theta_2^2 \\ \vdots \\ \theta_F^2 \end{bmatrix} + \dots + a_{jK}^X \begin{bmatrix} \theta_1^K \\ \theta_2^K \\ \vdots \\ \theta_F^K \end{bmatrix}$$

X-component of trajectory of j-th point as linear combination of K basis trajectories

X-component of trajectory of all point as linear combination of K basis trajectories

$$\begin{bmatrix} X_{11} & X_{12} & \dots & X_{1P} \\ X_{21} & X_{22} & \dots & X_{2P} \\ \vdots & \vdots & \ddots & \vdots \\ X_{F1} & X_{F2} & \dots & X_{FP} \end{bmatrix} = \begin{bmatrix} \theta_1^1 & \theta_2^1 & \dots & \theta_F^1 \\ \theta_1^2 & \theta_2^2 & \dots & \theta_F^2 \\ \vdots & \vdots & \ddots & \vdots \\ \theta_1^K & \theta_2^K & \dots & \theta_F^K \end{bmatrix} \begin{bmatrix} a_{11}^X & a_{12}^X & \dots & a_{1P}^X \\ a_{21}^X & a_{22}^X & \dots & a_{2P}^X \\ \vdots & \vdots & \ddots & \vdots \\ a_{K1}^X & a_{K2}^X & \dots & a_{KP}^X \end{bmatrix}$$

$$S^{X} = \Theta^X \times A^X$$

$F \times P \quad F \times K \quad K \times P$

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X, Y and Z-components of trajectory of all points

$$\begin{bmatrix} X_{11} & X_{12} & \dots & X_{1P} \\ X_{21} & X_{22} & \dots & X_{2P} \\ \vdots & \vdots & \ddots & \vdots \\ X_{F1} & X_{F2} & \dots & X_{FP} \end{bmatrix} = \begin{bmatrix} \theta_1^1 & \theta_2^1 & \dots & \theta_F^1 \\ \theta_1^2 & \theta_2^2 & \dots & \theta_F^2 \\ \vdots & \vdots & \ddots & \vdots \\ \theta_1^K & \theta_2^K & \dots & \theta_F^K \end{bmatrix} \begin{bmatrix} a_{11}^X & a_{12}^X & \dots & a_{1P}^X \\ a_{21}^X & a_{22}^X & \dots & a_{2P}^X \\ \vdots & \vdots & \ddots & \vdots \\ a_{K1}^X & a_{K2}^X & \dots & a_{KP}^X \end{bmatrix}$$

X, Y and Z-components of trajectory of all points

$$\begin{bmatrix} X_{11} & \dots & X_{1P} \\ Y_{11} & \dots & Y_{1P} \\ Z_{11} & \dots & Z_{1P} \\ \vdots & \vdots & \vdots \\ X_{F1} & \dots & X_{FP} \\ Y_{F1} & \dots & Y_{FP} \\ Z_{F1} & \dots & Z_{FP} \end{bmatrix} = \begin{bmatrix} \theta_1^1 & \dots & \theta_F^1 \\ \theta_1^2 & \dots & \theta_F^2 \\ \vdots & \vdots & \vdots \\ \theta_1^K & \dots & \theta_F^K \\ \theta_1^Y & \dots & \theta_F^Y \\ \theta_1^Z & \dots & \theta_F^Z \end{bmatrix} \begin{bmatrix} a_{11}^X & a_{11}^Y & a_{11}^Z & \dots & a_{1P}^X & a_{1P}^Y & a_{1P}^Z \\ a_{21}^X & a_{21}^Y & a_{21}^Z & \dots & a_{2P}^X & a_{2P}^Y & a_{2P}^Z \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ a_{K1}^X & a_{K1}^Y & a_{K1}^Z & \dots & a_{KP}^X & a_{KP}^Y & a_{KP}^Z \\ a_{11}^Y & a_{11}^Z & \dots & a_{1P}^Y & a_{1P}^Z & \dots & a_{1P}^X \\ a_{21}^Y & a_{21}^Z & \dots & a_{2P}^Y & a_{2P}^Z & \dots & a_{2P}^X \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{K1}^Y & a_{K1}^Z & \dots & a_{KP}^Y & a_{KP}^Z & \dots & a_{KP}^X \end{bmatrix} \begin{matrix} A^X \\ A^Y \\ A^Z \end{matrix}$$

$$S_{3F \times P} = \Theta_{3F \times 3K} A_{3K \times P}$$

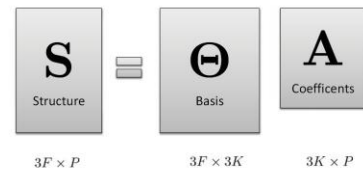
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TRAJECTORY REPRESENTATION

of Dynamic Structure



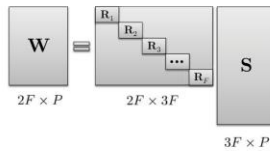
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TRAJECTORY REPRESENTATION

of Dynamic Structure Under Orthographic Projection



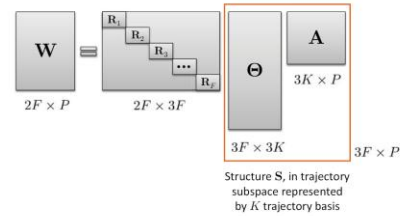
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TRAJECTORY REPRESENTATION

of Dynamic Structure Under Orthographic Projection



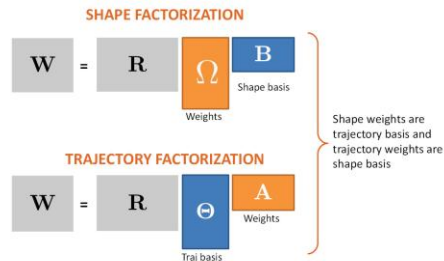
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DUALITY

Weights and Bases

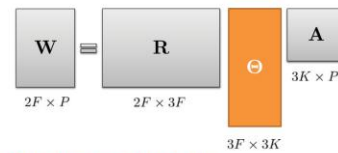


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ESTIMATING STRUCTURE VIA TRAJECTORY MODEL



Object Independent Basis

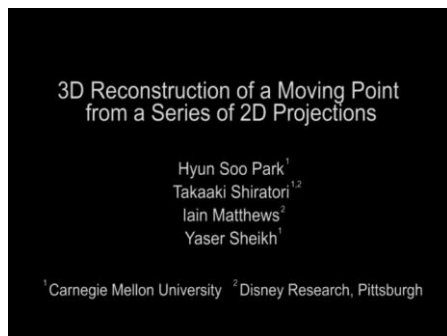
1. Deformation constrained by physical actuation
2. Trajectories vary smoothly and not randomly
3. Can be compactly represented by predefined basis e.g. Discrete Cosine Transform

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Results



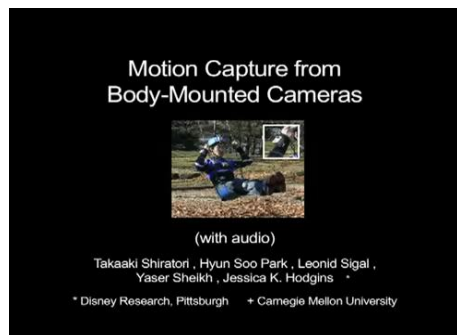
H. Park et al. 3D Reconstruction of a Moving Point from a Series of 2D Projections. ECCV'10.

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Structure from motion



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